# MUHAMMAD FADLURROHMAN

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## EDUCATION

**National Central University** **Taoyuan, Taiwan**
*Master of Science (M.Sc.)* Sep. 2023 - Jul. 2025

- **Advisor:** Prof. Jia-Ching Wang
- **Thesis:** Balanced-MixUp on Manifold for Highly Imbalanced Breast Cancer Classification
- **Coursework:** Introduction to Data Science, Introduction to Deep Learning, Practice in Audio/Video Signal Processing, Programming Language Design.
- **GPA:** 4.24/4.30

**Gadjah Mada University** **Yogyakarta, Indonesia**
*Master of Computer Science (M.Cs.)* Aug. 2022 – Jul. 2025

- **Advisor:** Afiahayati, S.Kom., M.Cs., Ph.D.
- **Thesis:** Balanced-MixUp on Manifold for Highly Imbalanced Breast Cancer Classification
- **Coursework:** Algorithm Analysis, Data Science, Data Warehouse and Intelligence Bussiness, Decision Support System, Digital Intelligence and Social Informatics, Information Management and Audit, Mathematics for Computer Science, Research Methodology for Computer Science.
- **GPA:** 3.88/4.00

**Yogyakarta University of Technology** **Yogyakarta, Indonesia**
*Bachelor of Computer (S.Kom.)* Sep. 2016 – Apr. 2021

- **Advisor:** Saucha Diwandari, S.Kom., M.Eng.
- **Thesis:** Expert System for Diagnosing Eye Diseases in Humans with Web-Based Forward Chaining Method
- **CGPA:** 3.50/4.00

## WORK EXPERIENCE

**Research Assistant** **Taoyuan, Taiwan**
*Deep Learning & Media System Lab, National Central University* Oct. 2023 – May. 2025

- Developed and trained machine learning and deep learning models for image classification and data analysis using Python, TensorFlow, and PyTorch.
- Built end-to-end ML pipelines including data cleaning, preprocessing, feature engineering, normalization, and data augmentation.
- Evaluated model performance using industry-standard metrics (accuracy, precision, recall, F1-score, ROC-AUC) to ensure model reliability.
- Performed hyperparameter tuning and model optimization to improve model accuracy and generalization.
- Analyzed model outputs and visualized results using confusion matrices and performance charts to support decision-making.
- Collaborated with team members to review results, improve model design, and document technical findings.

## PROJECTS

**Breast Lesion Detection & Classification on Mammograms** **Taoyuan, Taiwan**
*Deep Learning & Media System Lab, National Central University* Oct. 2023 – May. 2025

- Developed an object detection and classification pipeline to identify breast masses and calcifications (benign vs malignant) from mammogram images.
- Implemented bounding-box based lesion localization with multi-class labels (e.g., *MASS_BENIGN, MASS_MALIGNANT, CALC_BENIGN*).
- Applied image preprocessing and normalization to enhance lesion visibility and reduce background noise in mammography data.
- Trained and evaluated deep learning models for simultaneous detection and classification, supporting clinical decision-making workflows.
- Visualized prediction results with overlaid bounding boxes and class labels for model interpretability and validation.

## CONFERENCE

**Balanced-MixUp on Manifold for Highly Imbalanced Breast Cancer Classification** — **NTPU, Taiwan**
Muhammad Fadlurrohman, Farchan Hakim Raswa, Bach-Tung Pham, Afiahayati, Jia-Ching Wang — Aug. 2025
*The 38th Conference on Computer Vision, Graphics, and Image Processing (CVGIP2025)*

**Handling Imbalance with SMOTE Mixup on Breast Cancer Classification** — **NDHU, Taiwan**
Muhammad Fadlurrohman, Farchan Hakim Raswa, Bach-Tung Pham, Afiahayati, Jia-Ching Wang — Aug. 2024
*The 37th IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP2024)*

## PUBLICATION

**Benign Share Benefit to Malignant: Balanced-Mixing on Feature-Space for Imbalanced Breast Cancer Classification** — **Taoyuan, Taiwan**
Farchan Hakim Raswa, Muhammad Fadlurrohman, Bach-Tung Pham, Ika Candradewi, Afiahayati, Ming-Hsiang Su, Chung-I Huang, Kuo-Chen Li, Shih-Lun Chen, Yung-Hui Li, Jia-Ching Wang — Jun. 2026
*MDPI 2026*

**Balanced-MixUp on Manifold for Highly Imbalanced Breast Cancer Classification** — **Taoyuan, Taiwan**
Muhammad Fadlurrohman — Jul. 2025
*NCU 2025*

## ORGANIZATION

**Indonesian Students Association in Taiwan** — **Taipei, Taiwan**
*Department of Education* — Oct. 2023 – May. 2024
- Organized academic seminars, workshops, and discussions to support Indonesian students in Taiwan.
- Facilitated scholarship, research, and career development opportunities.

**Computer Science Graduate Student Association** — **Yogyakarta, Indonesia**
*Department of Public Relation & Information* — Apr. 2023 – Sep. 2023
- Developed and disseminated press releases, social media content, and newsletters to promote events.
- Managed official communication channels (e.g., Instagram, WhatsApp, website).
- Coordinated media coverage and interviews for departmental activities, including seminars, workshops, and competitions.
- Created visual and multimedia materials (e.g., posters, infographics, videos).

## AWARDS & SCHOLARSHIPS

**Dual Degree Scholarship to National Central University** — **Taoyuan, Taiwan**
*Awardee* — Sep 2023
- Selected for the dual degree scholarship, with full funding that covers tuition, fees, monthly stipend.
- Scholarship value: NTD ~106,480 per year for one years of master's study.

## SKILLS

**Programming:** Python (PyTorch, TensorFlow, Keras, YOLOv8, Scikit-learn, Numpy, Pandas), MATLAB, C, and SQL.
**Dev Tools**: Git, VS Code, Jupyter, Vim, and Claude Code.
**AI/ML**: Diffusion Models, LLMs, Computer Vision.

## LICENCES & CERTIFICATIONS

**IBM:** AI Developer Professional Certificate, AI Engineering Professional Certificate, Data Analyst Professional Certificate, Data Engineering Professional Certificate, Data Science Professional Certificate, Machine Learning Professional Certificate, RAG and Agentic AI Professional Certificate.
**Google:** Data Analytics Professional Certificate.
**Microsoft:** AI & ML Engineering Professional Certificate.

## LANGUAGES PROFICIENCY

**TOEFL ITP:** 597